

Score 1

Correct

Type : Mcq

Attempted

A retail Spark application has a pair RDD containing `(region, amount)` for every sale. The requirement is to calculate the total sales amount for each region. Which operation is suitable?

☐ `mapPartitions` without a key

☐ `collect()` before grouping

☐ `repartition(1)` only

☒ `reduceByKey(\_ + \_)`

✓ Correct Answer

Score 1

Correct

Type : Mcq

Attempted

A customer-registration system creates two 'Customer' objects with the same customer ID. The class correctly overrides 'equals()' and 'hashCode()', and both objects are added to a 'HashSet'. What should the set normally do?

- ☐ Keep both because equal hash codes always mean different objects
- ☐ Keep both because 'HashSet' always compares object references only
- ☐ Sort both automatically before checking equality

☒ Treat them as the same logical element when equality and hash code match

✓ Correct Answer



Search



Score

1

Correct

Type : Mcq

Attempted

A payment system has several unrelated classes that must all guarantee they provide the same set of operations, while each class can implement those operations differently. What is the main purpose of a Java interface in this design?

- ☐ Store every created object on heap memory
- ☐ Guarantee exactly one concrete implementation exists
- ☐ Replace all exception handling in an application
- ☒ Define a contract implementing classes must provide

✓ Correct Answer



Search



ENG
IN



20:
05-10-20

DELL

Score 1

Correct

Type: MCQ

Attempted

A live delivery dashboard calculates metrics in 10-minute windows. Business users still want events accepted when they arrive up to 2 minutes late because of network delay. Which combination expresses this requirement?

- ☒ Use event-time windows with a suitable watermark
- ☐ Use an event-time window but no watermark for late-data control
- ☐ Use only a processing-time window with no event-time handling
- ☐ Use a watermark without defining the required time window

Correct Answer



Search



ENG

IN



2023

05-10-2023

DELL

Score

0

Incorrect

Type : Mcq

Attempted

A file-processing service opens a report file, reads its contents, and may throw an exception while processing. The developer wants the file resource to be closed reliably without writing repetitive cleanup code. Why is try-with-resources useful?

☐ It guarantees faster file access automatically

☒ It closes eligible file resources automatically

☐ It removes checked file exceptions automatically

☐ It converts files into memory objects

✓ Correct Answer

✗ Incorrect Answer



Q Search



Score 1

Correct

Type : Mcq

Attempted

A partner system drops new CSV files into a landing directory throughout the business day. The data engineering team wants Spark to continuously process newly arriving files as streaming input. What can Structured Streaming do?

☐ Read the directory only once and ignore files that arrive later

☒ Treat newly arriving files as incremental streaming input

✓ Correct Answer

☐ Require a socket source before any file can be processed

☐ Support only Parquet files and not CSV file streams



Search



ENG
IN



2024
05-10-2026

Score

1

Correct

Type : Mcq

Attempted

A banking program updates both an accounts table and an audit table as one business operation. If either update fails, neither table should retain the change. Which JDBC approach is appropriate?

- ☒ Disable auto-commit, run both updates, then commit or rollback
- ☐ Skip exception handling and commit each statement separately
- ☐ Commit after updating the first table, then continue
- ☐ Use separate connections with auto-commit enabled for each

✓ Correct Answer



Search



ENG
IN



20
05-10-20

Score 0

Incorrect

Type : Mcq

Attempted

A stateful Structured Streaming application processes account events. After a driver failure, the application must restart and continue using the previously stored progress and state instead of starting from the beginning. What should be configured?

☐ Configure a durable checkpoint location

✓ Correct Answer

☒ Cache the query result only in executor memory

✗ Incorrect Answer

☐ Repartition the streaming DataFrame on every trigger

☐ Create a temporary SQL view for the stream



Search



ENG
IN



2024

05-10-2026

DELL

Score

1

Correct

Type : Mcq

Attempted

A data engineer is processing structured order data and needs named columns, SQL expressions, schema awareness, and Spark optimizer support. Which Spark abstraction is more suitable?

- ☐ A text-file iterator with no structured column metadata
- ☐ A local Scala collection processed only on the driver
- ☐ An RDD using low-level functions on unstructured objects
- ☒ A DataFrame with a defined schema and SQL expressions

✓ Correct Answer



Search



ENG
IN



20:24

05-10-2026

DELL

Score 1

Correct

Type : Mcq

Attempted

Two threads in an order service both increment the same shared counter. After heavy concurrent activity, the final count is lower than expected because some updates were overwritten. Which concept best explains the problem?

- ☐ A livelock where threads keep reacting without making progress
- ☐ A deadlock between two locks
- ☒ A race condition caused by unsynchronized shared updates
- ☐ Thread starvation caused by scheduling priority

✓ Correct Answer



Search



ENG
IN



20:24

05-10-2026

DELL

Score 1

Correct

Type : Mcq

Attempted

Two worker threads process inventory updates. Thread A already holds Lock 1 and is waiting for Lock 2, while Thread B holds Lock 2 and is waiting for Lock 1. Neither thread can proceed. What concurrency problem has occurred?

☒ A deadlock

✓ Correct Answer

☐ Thread starvation

☐ A livelock

☐ A race condition



Search



ENG
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05-10

Score

1

Correct

Type : Mcq

Attempted

A Spark transformation can be implemented either with an existing built-in string function or with a newly written Python/Scala UDF. The team wants the option that Spark can optimize more effectively. Why is the built-in function usually preferred?

☒ Spark can understand and optimize built-in expressions more easily

✓ Correct Answer

☐ A built-in function automatically removes all null values

☐ A built-in function guarantees that no shuffle can occur

☐ A built-in function always executes only once per job



Search

ENG
IN

05

Score 1

Correct

Type : Mcq

Attempted

A Scala service classifies transaction-risk scores. A score of 5 is evaluated using `5 match { case x if x > 3 => "high"; case _ => "low" }`. What result should the service produce?

☐ It returns the value 'low'

☐ It returns the number '5'

☒ It returns the value 'high'

☐ It causes a compilation failure

✓ Correct Answer



Q Search



Score 1

Correct

Type : Mcq

Attempted

A document service must copy customer PDF files without changing any bytes in the file. Which Java I/O family is appropriate for this binary-copy requirement?

- ☐ `BufferedReader` and `PrintWriter` text streams
- ☐ `Scanner` and `Formatter` token/text utilities
- ☒ `FileInputStream` and `FileOutputStream` byte streams
- ☐ `FileReader` and `FileWriter` character streams

✓ Correct Answer



Q Search

ENG
IN20:25
05-10-2026

Score 0

Incorrect

Type : Mcq

Attempted

An e-commerce application has a stream of product prices. It must keep only prices above 500 and then double each remaining value for a simulation. Which Java stream expression performs the requirement?

☒ `prices.stream().filter(p → p > 500).collect().map(p → p * 2)`

✗ Incorrect Answer

☐ `prices.stream().map(p → p * 2).filter(p → p > 500)`

☐ `prices.stream().collect(p → p > 500).map(p → p * 2)`

☐ `prices.stream().filter(p → p > 500).map(p → p * 2)`

✓ Correct Answer



Search



ENG
IN



20:26
05-10-2026

Score 1

Correct

Type : Mcq

Attempted

An HR application stores employee IDs in the exact order they were entered, and the same ID may appear more than once during testing. Which Java collection interface is the best starting choice?

☐ Set☐ Map☒ List☐ Queue only

✓ Correct Answer



Search

20:26
05-10-2026

Score 0

Incorrect

Type : Mcq

Attempted

A multi-threaded Java service stores session information in a shared map that is updated by several threads at the same time. At a conceptual level, why might 'ConcurrentHashMap' be chosen over 'HashMap'?

- ☐ It guarantees all keys remain sorted automatically
- ☒ It automatically opens and manages database connections
- ☐ It supports safer access from multiple concurrent threads
- ☐ It prevents every possible application concurrency bug

✗ Incorrect Answer

✓ Correct Answer



Search



ENG
IN



Score

1

Correct

Type : Mcq

Attempted

A JDBC money-transfer operation successfully debits the source account, but the subsequent credit to the destination account fails. The business rule requires the transfer to be all-or-nothing. What should the program do?

- ☐ Ignore the failed transaction and retry only the credit separately
- ☐ Commit the debit even though the credit failed
- ☐ Enable auto-commit and continue with the partial update

☒ Roll back the entire database transaction

✓ Correct Answer



Search



ENG
IN



20:27
05-10-2026

DELL

Score 0

Incorrect

Type : Mcq

Attempted

A team is starting a new Spark real-time project and must choose between the older DStream API and the newer Structured Streaming model. Which statement is most appropriate for the new implementation?

☐ Structured Streaming cannot support event-time operations

☐ Structured Streaming is the newer DataFrame-based streaming model

☐ DStreams are the newer DataFrame-based streaming model

☒ DStreams and Structured Streaming expose exactly the same programming model

✓ Correct Answer

✗ Incorrect Answer



Search



ENG
IN



20.2
05-10-2021

DELL

Score 1

Correct

Type : Mcq

Attempted

A transport application declares a 'Vehicle' reference, but at runtime it points to a 'Car' object. The 'Car' class overrides 'move()'. When the application calls 'move()' through the 'Vehicle' reference, which method runs?

☒ The overridden 'Car' implementation runs

✓ Correct Answer

☐ No implementation runs through the reference

☐ The inherited 'Vehicle' implementation runs

☐ Both implementations run automatically together



Search



ENG
IN



20:27
05-10-2026

Score 1

Correct

Type : Mcq

Attempted

A trainee is building a score-processing program and wants the compiler to prevent non-integer values from being inserted into the score collection. Why is 'List' safer than using a raw 'List'?

- ☐ It makes stored list elements immutable
- ☐ It sorts stored scores automatically
- ☐ It prevents duplicate scores automatically
- ☒ It provides compile-time element type checking

 Correct Answer

Q Search



2027

05-10-2026

DELL

Score

1

Correct

Type : Mcq

Attempted

A data engineer notices that a Spark DataFrame query is taking longer than expected. Before changing the code, the engineer wants to inspect how Spark plans to execute the query. Which tool is the best first check?

☐ Call `df.show()` to display sample rows

☒ Call `df.explain()` to inspect the plan

☐ Call `df.cache()` to mark the DataFrame for reuse

☐ Call `df.printSchema()` to inspect column types

✓ Correct Answer



Search



ENG
IN



20:
05-10-20

DELL

Score

1

Correct

Type : Mcq

Attempted

In a payroll application, the 'Manager' constructor must send the employee ID to the 'Employee' parent constructor before initializing manager-specific fields. Which call should the child constructor use?

☐ 'super()' calls only the parent no-argument constructor

☒ 'super(value)' passes the value to a parent constructor

☐ 'this(value)' calls another constructor in the same class

☐ 'this()' calls the current class no-argument constructor

✓ Correct Answer



Search

ENG
IN202
05-10-202

Score 1

Correct

Type : Mcq

Attempted

A Java application starts a time-consuming remote calculation but should continue doing other work and process the result later when it becomes available. What does 'CompletableFuture' help represent?

- ☐ It represents a file lock held permanently
- ☒ It represents an asynchronous result completed later
- ☐ It represents the JVM garbage collection process
- ☐ It represents a database table schema definition

✓ Correct Answer



Search



DELL



Score 0

Incorrect

Type : Mcq

Attempted

A developer compiles a Java application on a Windows laptop, then the deployment team copies only the compiled `.class` file to a Linux server. Which component allows the same bytecode to execute on both operating systems?

☒ The JAR filename on each system

✗ Incorrect Answer

☐ The source editor on each system

☒ The JVM available on each system

✓ Correct Answer

☐ The package statement on each system



Search

ENG
IN202
05-10-202

DELL

Score 1

Correct

Type : Mcq

Attempted

Two regional teams deliver CSV files for the same sales dataset, but schema inference reads one numeric column differently in the two files. Before the files are combined, what should the team do?

- ☐ Union first and resolve incompatible column types only after failure
- ☐ Convert every column in both files to String regardless of meaning
- ☐ Infer each file independently and union them without checking types

☒ Apply the same explicit schema when reading both files

✓ Correct Answer



Search



ENG
IN



2028
05-10-2026

Score 0

Incorrect

Type : Mcq

Attempted

During a Spark job, one executor loses an RDD partition. Spark needs to recreate that partition instead of recomputing the entire application manually. Which information helps Spark rebuild it?

- ☐ Only the notebook cell that first displayed the RDD
- ☒ Only the executor memory that previously held the partition
- ☐ The RDD lineage of transformations
- ☐ Only the name of a temporary SQL view

✗ Incorrect Answer

✓ Correct Answer



Search



ENG
IN



05

DELL

Score

1

Correct

Type : Mcq

Attempted

A developer is modelling a 'Car' that uses an 'Engine' object as one of its internal parts rather than making 'Car' a subtype of 'Engine'. Which code design most clearly represents composition?

☐ 'Order' implements a 'Payable' interface

☒ 'Order' contains a 'Payment' object as a field

☐ 'Order' extends 'Payment' to inherit its behavior

☐ 'Payment' extends 'Order' to reuse order behavior

✓ Correct Answer



Search



ENG
IN



05-10-20

DELL

Score 1

Correct

Type : Mcq

Attempted

A Spark job has an RDD of sales records and the developer wants to define another RDD by changing each record, without immediately producing a final result. Which Spark RDD operation is a transformation?

☐ `collect()`

☒ `filter()`

☐ `first()`

☐ `count()`

✓ Correct Answer



Search



ENG
IN



20
05-10-20

Score 1

Correct

Type : Mcq

Attempted

A banking application must prevent other classes from directly changing an account balance and should expose controlled methods for updates. Which class design best demonstrates encapsulation?

- ☐ Expose every field publicly to all callers
- ☐ Allow callers to modify internal arrays directly
- ☐ Store unrelated data inside global variables directly
- ☒ Keep fields private and expose controlled methods

✓ Correct Answer



Search



ENG



04-10